

CASE RECORDS of the MASSACHUSETTS GENERAL HOSPITAL

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Case 23-2021: A 41-Year-Old Woman with Bloody Stools and Thrombocytopenia

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PRESENTATION OF CASE

Dr. Matthew B. Roberts: A 41-year-old woman who had undergone renal transplantation was admitted to this hospital because of bloody stools and thrombocytopenia.

The patient had been in her usual state of health until 3 weeks before this admission, when mild rectal discomfort and nonbloody diarrhea developed. The episodes of diarrhea occurred several times a day and were not associated with fever, chills, abdominal pain, nausea, or vomiting.

Two weeks before this admission, the patient sought evaluation at this hospital. The abdominal and rectal examinations were normal. The hemoglobin level was 11.0 g per deciliter (reference range, 14.0 to 18.0) and the platelet count 129,000 per microliter (reference range, 150,000 to 450,000). Stool cultures and tests of the stool for *Clostridioides difficile* antigen and toxin and Shiga toxin were negative, as was examination of the stool for ova and parasites. Other laboratory test results are shown in Table 1. Treatment with docusate was stopped, treatment with topical rectal hydrocortisone cream was started, and a recommendation was made to increase the amount of fiber in the diet. During the next week, the episodes of diarrhea decreased in frequency to once a day; however, the rectal pain persisted.

One week before this admission, the patient was evaluated by her primary care physician because the rectal pain had worsened; she was unable to sit down, and new tenesmus was present. A rectal examination revealed mild tenderness and thin yellow-brown liquid stool, with no evidence of blood or an external anal injury. A radiograph of the abdomen was normal. Docusate, senna, and topical lidocaine jelly were prescribed.

Three days before this admission, bowel movements with tarry black and loose maroon stools and dark clots began to occur multiple times a day. When the episodes increased in frequency to every hour and there was bright-red blood per rectum, the patient presented to the emergency department of this hospital for evaluation.

In the emergency department, the patient reported severe rectal pain and burn-

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Table 1. Laboratory Data.*

Variable	Reference Range, This Hospital†	2 Wk before Admission	On Admission
Blood			
Platelet count (per μ l)	150,000–450,000	129,000	0
White-cell count (per μ l)	4500–11,000	7500	8490
Differential count (per μ l)			
Neutrophils	1800–7700	6428	6920
Lymphocytes	1000–4800	255	1120
Monocytes	20–120	803	300
Eosinophils	0–900	8	0
Basophils	0–300	8	80
Hemoglobin (g/dl)	14.0–18.0	11.0	6.0
Hematocrit (%)	42.0–52.0	32.9	19.6
Mean corpuscular volume (fl)	80.0–100.0	—	74.5
Mean corpuscular hemoglobin (pg)	26.0–34.0	—	22.8
Red-cell distribution width (%)	11.5–14.5	—	14.8
Reticulocytes (%)	0.7–2.5	—	1.3
Alanine aminotransferase (U/liter)	0–41	—	80
Aspartate aminotransferase (U/liter)	10–50	—	104
Albumin (g/dl)	3.5–5.2	—	2.5
Total bilirubin (mg/dl)	0.0–1.0	—	0.5
Lactate dehydrogenase (U/liter)	110–210	—	465
Haptoglobin (mg/dl)	30–200	—	38
Prothrombin time (sec)	11.5–14.5	—	13.9
International normalized ratio	0.9–1.1	—	1.1
Activated partial-thromboplastin time (sec)	22.0–36.0	—	33.2
Fibrinogen (mg/dl)	150–400	—	288
D-Dimer (ng/ml)	<500	—	1650
Creatinine (mg/dl)	0.60–1.50	0.86	0.85
Urea nitrogen (mg/dl)	8–25	11	12
Potassium (mmol/liter)	3.4–5.0	3.7	4.2
Sodium (mmol/liter)	135–145	135	132
Glucose (mg/dl)	70–110	196	234
Urine			
Bilirubin	Negative	—	Negative
Urobilinogen	Negative	—	Negative
Blood	Negative	—	2+
Glucose	Negative	—	1+
Ketones	Negative	—	1+
Leukocyte esterase	Negative	—	Negative
Nitrites	Negative	—	Negative

* To convert the values for bilirubin to micromoles per liter, multiply by 17.1. To convert the values for creatinine to micromoles per liter, multiply by 88.4. To convert the values for urea nitrogen to millimoles per liter, multiply by 0.357. To convert the values for glucose to millimoles per liter, multiply by 0.05551.

† Reference values are affected by many variables, including the patient population and the laboratory methods used. The ranges used at Massachusetts General Hospital are for adults who are not pregnant and do not have medical conditions that could affect the results. They may therefore not be appropriate for all patients.

ing, as well as fatigue, dizziness, and weakness. She did not report fever, chills, abdominal pain, hematemesis, hemoptysis, epistaxis, rash, bruising, hematuria, or menorrhagia.

The patient was congenitally deaf. She had received a diagnosis of kidney disease 12 years earlier, when proteinuria and hypertension developed during pregnancy and did not resolve postpartum. Pathological examination of a renal biopsy specimen had reportedly revealed focal segmental glomerulosclerosis. The patient was presumed to have Alport's syndrome; no genetic testing had been performed.

Ten years before the current evaluation, the patient had discontinued all prescribed medications, including prednisone and labetalol, without talking to her doctors; 3 years later, she presented to this hospital with dyspnea on exertion and end-stage renal failure. Hemodialysis was started and continued until a deceased-donor renal transplantation was performed 6 months before the current evaluation.

Routine pretransplantation serologic testing of the patient revealed IgG antibodies to Epstein-Barr virus (EBV) and no antibodies to cytomegalovirus (CMV). Serologic testing of the donor also revealed IgG antibodies to EBV and no antibodies to CMV. Induction immunosuppressive therapy with antithymocyte globulin, belatacept, prednisone, and everolimus was initiated; famciclovir was administered for 3 months after transplantation. Treatment with prednisone was stopped 2 months after transplantation, when diabetes developed.

At the time of the current evaluation, medications included aspirin, linagliptin, trimethoprim-sulfamethoxazole, and omeprazole, as well as cholecalciferol, docusate, senna, topical hydrocortisone cream, and topical lidocaine jelly. The dose of everolimus had been decreased 1 month before this evaluation; belatacept had last been infused 2 weeks earlier. There were no known drug allergies.

The patient lived with her male partner and young son in an urban area of New England. She had not traveled recently. She had previously worked part-time as a teacher. There had been no exposure to animals other than two pet dogs. She had never smoked tobacco and did not drink alcohol. Her sister was deaf and had a history of renal failure that had led to renal transplanta-

tion. Her father had a history of diabetes and renal failure that required hemodialysis.

On examination, the temperature was 37.2°C, the blood pressure 85/46 mm Hg, the pulse 130 beats per minute, the respiratory rate 16 breaths per minute, and the oxygen saturation 100% while the patient was breathing ambient air. She appeared pale and unwell. There was no jaundice or lymphadenopathy. There were palatal petechiae and sublingual ecchymoses, but there was no bleeding from the gums or nose. There was a healed surgical incision in the right lower abdomen and no tenderness over the renal allograft. A rectal examination revealed tenderness and dark blood with clots.

The hemoglobin level was 6.0 g per deciliter and the platelet count 0 per microliter. A peripheral-blood smear showed red cells that were generally normal in size and morphology, with few hypochromic cells and rare pencil cells; there were no schistocytes or platelets. A direct antiglobulin test was negative. Blood cultures were obtained. Other laboratory test results are shown in Table 1.

Dr. Onofrio A. Catalano: Computed tomography (CT) of the abdomen and pelvis, performed without the administration of intravenous contrast material, revealed no bowel-wall thickening or distention. The spleen and liver were normal in size. The renal allograft was visible in the right lower quadrant (Fig. 1). The transplanted kidney was enlarged and surrounded by extensive perinephric fat stranding. A small amount of free fluid was also present. There was moderate hydronephrosis that extended up to the ureterovesical anastomosis. The renal collecting system, pelvis, and ureter contained spontaneously hyperdense material that was suggestive of blood breakdown products. No obstructing calculi were seen.

Dr. Roberts: Intravenous fluids were administered, and the blood pressure increased to 97/53 mm Hg. The patient was admitted to the hospital, and treatment with everolimus and trimethoprim-sulfamethoxazole was stopped. During the next 24 hours, 4 units of packed red cells and 5 units of pooled platelets were transfused. On the second hospital day, the hemoglobin level was 11.0 g per deciliter and the platelet count 4 per microliter.

A diagnostic test was performed.

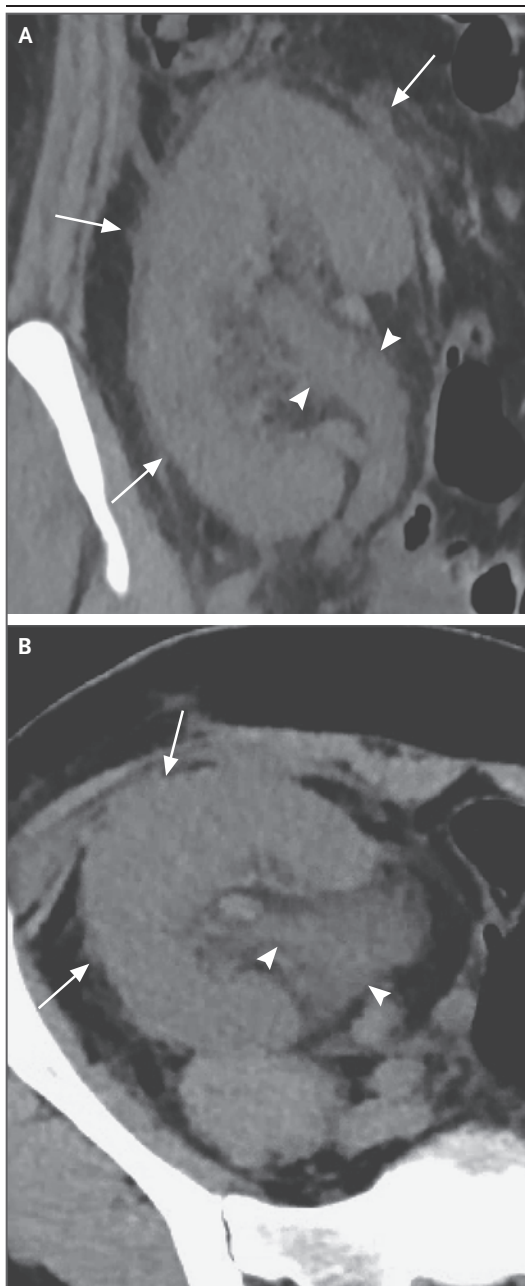


Figure 1. CT of the Renal Allograft.

Coronal and axial images (Panels A and B, respectively), obtained without the administration of intravenous contrast material, show a renal allograft in the right lower quadrant. The transplanted kidney is enlarged and surrounded by extensive perinephric fat stranding (arrows). A small amount of free fluid is also present. There is moderate hydronephrosis. The renal collecting system, pelvis, and ureter contain spontaneously hyperdense material (arrowheads) that is suggestive of blood breakdown products. No obstructing calculi are shown.

DIFFERENTIAL DIAGNOSIS

Dr. Olivia C. Smibert: This 41-year-old recipient of a deceased-donor kidney transplant presented with a biphasic illness, characterized initially by proctocolitis and subsequently by profound thrombocytopenia and gastrointestinal hemorrhage, features consistent with immune thrombocytopenia. Proctocolitis and immune thrombocytopenia are typically independent processes, but since these two conditions developed during the same time period, I will build my differential diagnosis around these disease processes in an attempt to establish a unifying diagnosis. Review of the patient's immunosuppressive regimen, the results of peritransplantation screening of the donor and recipient, and the patient's history of epidemiologic exposures may provide important clues in determining the most likely cause of this presentation.

PROCTOCOLITIS

This patient's diarrhea, rectal pain, and tenesmus suggest an abnormality that involves both the colon and the rectum. On the basis of the clinical history, laboratory findings, and imaging studies provided, several important noninfectious causes of proctocolitis — including cancer, ischemic colitis, and inflammatory bowel disease — can be ruled out.

The absence of a mass on rectal examination and the absence of signs of a regional or distant tumor on abdominal CT make primary bowel cancer unlikely. EBV-induced post-transplantation lymphoproliferative disorder can involve the gastrointestinal tract and cause gastrointestinal bleeding, but this condition is often associated with constitutional symptoms and occurs much less frequently in EBV-seropositive recipients than in EBV-seronegative recipients. In addition, immunosuppression with mammalian target of rapamycin (mTOR) inhibitors has been associated with reduced carcinogenesis after single-organ transplantation.

Intestinal ischemia is typically associated with a more acute onset than that observed in this patient, and abdominal pain is the predominant symptom. New diagnoses of inflammatory bowel disease after single-organ transplantation have occasionally been described; the condition can cause bloody diarrhea, but it typically occurs years rather than months after transplantation,

which is not consistent with this patient's presentation.

Infections with *C. difficile*, shigella, and campylobacter species are unlikely in this patient because of the negative stool culture. Hemolytic-uremic syndrome associated with Shiga toxin-producing *Escherichia coli* or *Shigella dysenteriae* is typically a biphasic illness, characterized initially by abdominal pain, vomiting, and diarrhea and subsequently by thrombocytopenia, hemolytic anemia, and acute kidney injury. Although this patient's colitis with hemorrhage and thrombocytopenia is consistent with this diagnosis, the findings on examination of a peripheral-blood smear and laboratory evaluation are not consistent with microangiopathic hemolysis or acute kidney injury associated with this syndrome. Furthermore, the negative immunoassay for Shiga toxin makes this diagnosis unlikely.

Hemorrhagic gastroenteritis is infrequently caused by infection with *Vibrio parahaemolyticus*, which is transmitted by ingestion of contaminated seafood and is occasionally associated with disseminated intravascular coagulation. This diagnosis is unlikely, since the median duration of illness is days rather than weeks and the patient did not have a history of recent seafood ingestion. She presented with hypotension and bleeding, but the absence of coagulopathy on laboratory evaluation is not consistent with disseminated intravascular coagulation. A stool examination was negative for ova, cysts, and parasites, and the features of her presentation would be atypical for a fungal infection. The remaining considerations in this case include sexually transmitted infections associated with proctitis, as well as infections caused by the herpesvirus family of pathogens.

SEXUALLY TRANSMITTED INFECTIONS

Gastrointestinal manifestations of common sexually transmitted infections are not always recognized by clinicians.¹ Although a more detailed history was not available for this patient, a detailed sexual history should be considered in the evaluation of any patient who presents with rectal symptoms. Infections with *Neisseria gonorrhoeae*, *Chlamydia trachomatis*, and *Treponema pallidum* can all cause a spectrum of rectal symptoms, including typical features of proctitis, and severe disease can occasionally mimic inflammatory bowel

disease. However, infections with these organisms are not typically associated with pathological findings proximal to the rectum or with extraintestinal hematologic manifestations.^{2,3} The most invasive serotypes of *C. trachomatis* can cause lymphogranuloma venereum, with a hemorrhagic proctocolitis that is typically associated with severe anorectal pain. However, lymphogranuloma venereum often leads to painful lymphadenopathy (although this finding is less common in women than in men) and is rarely reported in the United States outside of an outbreak.^{4,5}

Herpes simplex virus can also cause painful proctitis, often with associated perianal lesions. Invasive disease may involve the colon proximal to the rectum, particularly in immunocompromised patients.⁶ In the presence of immunocompromise, reactivation of latent virus is more likely than primary infection. Although proctocolitis due to herpes simplex virus has been associated with extraintestinal manifestations, including sacral anesthesia, urinary retention, and impotence, it is not frequently associated with immune thrombocytopenia or other hematologic abnormalities.^{2,6-9} Therefore, herpes simplex virus infection would be a reasonable explanation of the proctocolitis in this patient, but the absence of perianal lesions on external rectal examination and the presence of profound hematologic manifestations would be atypical of this diagnosis.

CYTOMEGALOVIRUS

CMV is a notable cause of post-transplantation complications and death. Although both the donor and the recipient were seronegative for CMV in this case, the patient remains at risk for acquired primary CMV infection, which is more severe in immunocompromised patients.^{3,10} Transmission of CMV can occur through multiple routes, including sexual intercourse, close contacts, and occupational exposures, such as child care.¹¹⁻¹³ This patient had a number of risk factors for primary acquisition of CMV, including a child at home and a previous occupation with exposure to children. Tissue-invasive CMV infection frequently involves the gastrointestinal tract, and proctocolitis has been well described with both primary infection and reactivation of latent disease.¹⁴⁻¹⁶ CMV infection is also associated with a range of hematologic manifestations, including anemia, the development of cold agglutinins,

disseminated intravascular coagulation, and immune thrombocytopenia.¹⁷⁻¹⁹

IMMUNE THROMBOCYTOPENIA

Immune thrombocytopenia is an acquired thrombocytopenia caused by antiplatelet autoantibodies directed against platelet antigens.²⁰ Molecular mimicry between viral antigens and host proteins has been implicated in the pathogenesis of secondary immune thrombocytopenia, which may be associated with several acute and chronic infections.²⁰ Acute immune thrombocytopenia has most frequently been observed in children, often after a recent documented illness, with CMV and other members of the herpesvirus family frequently implicated.²¹ CMV-associated immune thrombocytopenia has been described in recipients of solid-organ transplants and is more likely to be associated with primary infection than with reactivation of latent disease. CMV-associated immune thrombocytopenia may be refractory to standard therapy; there are some reports of paradoxical worsening of thrombocytopenia in the presence of glucocorticoids and improvement in response to antiviral therapy.^{17,22-25}

IMMUNOSUPPRESSION WITH BELATACEPT

This patient had received belatacept, a fusion protein composed of the Fc fragment of human IgG1 linked to the extracellular domain of cytotoxic T-lymphocyte-associated antigen 4, which was developed as a calcineurin inhibitor-sparing agent for use in kidney-transplant recipients.²⁶ Because calcineurin inhibitors act downstream of the T-cell receptor, they inhibit activation of both mature and immature T cells; by contrast, the effect of belatacept is decreasingly evident in increasingly mature T cells.²⁷ The side-effect profile of belatacept includes impaired antiviral immunity, particularly in patients who have a primary viral infection and have not been previously infected with the virus (Fig. 2).²⁷ This effect of belatacept was first observed in EBV-seronegative kidney-transplant recipients, in whom the risk of EBV-induced post-transplantation lymphoproliferative disorder was 9 times the risk in EBV-seropositive recipients. The Food and Drug Administration has issued a warning against the use of belatacept in EBV-seronegative patients.

With increasing use of belatacept, a growing number of reports describe atypical and severe

CMV infection in the context of kidney transplantation.²⁶ These reports suggest higher rates of primary CMV disease, specifically severe, tissue-invasive, and life-threatening disease, and higher rates of resistance to ganciclovir in recipients treated with belatacept than in those treated with calcineurin inhibitors.²⁸⁻³⁰

This patient was seronegative for CMV at the time of transplantation, which put her at high risk for severe and uncontrolled primary viral infection in the presence of belatacept immunosuppression, because of the inhibition of critical T-cell antiviral immune responses to new infection. CMV disease frequently affects the gastrointestinal tract, including the squamous epithelium of the rectum, where there is a high density of somatic nerve endings that may be associated with clinically significant pain. CMV is also well described to be associated with the production of antiplatelet autoantibodies, and immune thrombocytopenia can occur during primary infection. The constellation of symptoms and laboratory findings in this case is highly suggestive of severe primary CMV infection causing proctocolitis and secondary immune thrombocytopenia in the context of impaired antiviral immunity to new infection due to belatacept immunosuppression. To establish this diagnosis, I would obtain blood samples for a CMV nucleic acid test; if the test is negative, I would consider sigmoidoscopy with biopsy.

DR. OLIVIA C. SMIBERT'S DIAGNOSIS

Primary cytomegalovirus infection associated with proctocolitis in the context of belatacept immunosuppression, complicated by gastrointestinal bleeding due to secondary immune thrombocytopenia.

DIAGNOSTIC TESTING

Dr. Roberts: This patient was seronegative for CMV before transplantation, and she received a kidney from a CMV-seronegative donor, so the risk of donor-derived CMV infection was very low. Primary CMV infection could have resulted from a close contact, or the patient could have had a false negative serologic test before transplantation. She lived with a young child and had been exposed to children through her work. The

diagnosis of CMV infection was established by testing of a blood sample for CMV DNA, which was detected at a level of 883,000 IU per milliliter.

LABORATORY DIAGNOSIS

Primary cytomegalovirus infection.

DISCUSSION OF MANAGEMENT

MANAGEMENT OF IMMUNE THROMBOCYTOPENIA

Dr. Katayoon Goodarzi: After transfusion of 12 units of platelets, the patient's platelet count increased from 0 to 4 per microliter. On the basis of this poor transfusion response, we suspected peripheral destruction of platelets, which supported a tentative diagnosis of virus-associated immune thrombocytopenia.³¹ Her rectal bleeding remained active, with platelet counts persistently lower than 10 per microliter. In order to rapidly increase her platelet count and prevent clinically significant blood loss, we decided to administer glucocorticoids and intravenous immune globulin (IVIG), which are standard treatments for patients with immune thrombocytopenia.³² After discussion with the primary care and infectious disease teams, we initiated therapy with prednisone (1 mg per kilogram of body weight per day), in addition to providing two doses of IVIG. The platelet count transiently increased to 47 per microliter but then decreased to 4 per microliter shortly thereafter, raising concerns about treatment-refractory disease. At this time, a CMV nucleic acid test showed a viral load of 883,000 IU per milliliter, a finding that corroborated the diagnosis of CMV-related immune thrombocytopenia.

While the infectious disease team administered antiviral therapy, we elected to administer a romiplostim injection on a weekly basis to enhance bone marrow thrombopoiesis. By the fourth hospital day, the patient's platelet count had risen above 10 per microliter.

MANAGEMENT OF CYTOMEGALOVIRUS INFECTION

Dr. Roberts: The initial management of CMV disease in solid-organ transplant recipients includes the institution of antiviral therapy in conjunction with weekly monitoring by means of CMV nucleic acid tests performed on the same sample type (whole blood or plasma) at the same laboratory.¹⁰ First-line antiviral options include oral

valganciclovir and intravenous ganciclovir.¹⁰ In patients with renal failure, dosing of these agents requires adjustment for the calculated creatinine clearance to reduce the incidence of side effects, as well as avoidance of inappropriately low doses, which may increase the risk of CMV that is refractory or resistant to treatment.^{10,33} Guidelines recommend the administration of antiviral therapy at therapeutic dosage for a minimum of 2 weeks, until the patient has clinically recovered and two consecutive measurements of the CMV DNA viral load separated by 1 week indicate CMV DNA eradication.¹⁰ This patient had severe disease complicated by a rare but serious manifestation of immune thrombocytopenia and had a very high CMV DNA viral load. Therefore, we elected to administer intravenous ganciclovir as the initial therapy.

In the absence of antiviral therapy, control of CMV replication is maintained only by the host immune response, so a reduction in immunosuppression may be considered in solid-organ transplant recipients with CMV infection who do not have concomitant allograft rejection.¹⁰ However, the benefit of reducing immunosuppression must be cautiously balanced against the risk of precipitating allograft rejection, which may in fact be increased by the presence of CMV infection; therefore, this approach is not appropriate in all cases.^{10,34} The patient was treated with high-dose prednisone for immune thrombocytopenia, and other immunosuppression was reduced accordingly, with the intention of maintaining an overall similar degree of immunosuppression.

CMV hyperimmunoglobulin therapy is a possible therapeutic adjunct, although its role is not clearly established.¹⁰ This patient received IVIG as part of treatment for immune thrombocytopenia; given the high seroprevalence of CMV in the community of blood donors, passive transfer of CMV immunoglobulin was likely.³⁵ The specific therapeutic effect of IVIG for the treatment of CMV infection in this patient is unknown.

After 1 week of antiviral therapy, the diarrhea and bleeding had resolved. The patient was discharged home on the ninth hospital day with a platelet count of 129,000 per microliter and a CMV DNA viral load of 83,300 IU per milliliter. Trials have shown the noninferiority of oral valganciclovir to intravenous ganciclovir among patients with nonsevere CMV infection who have good

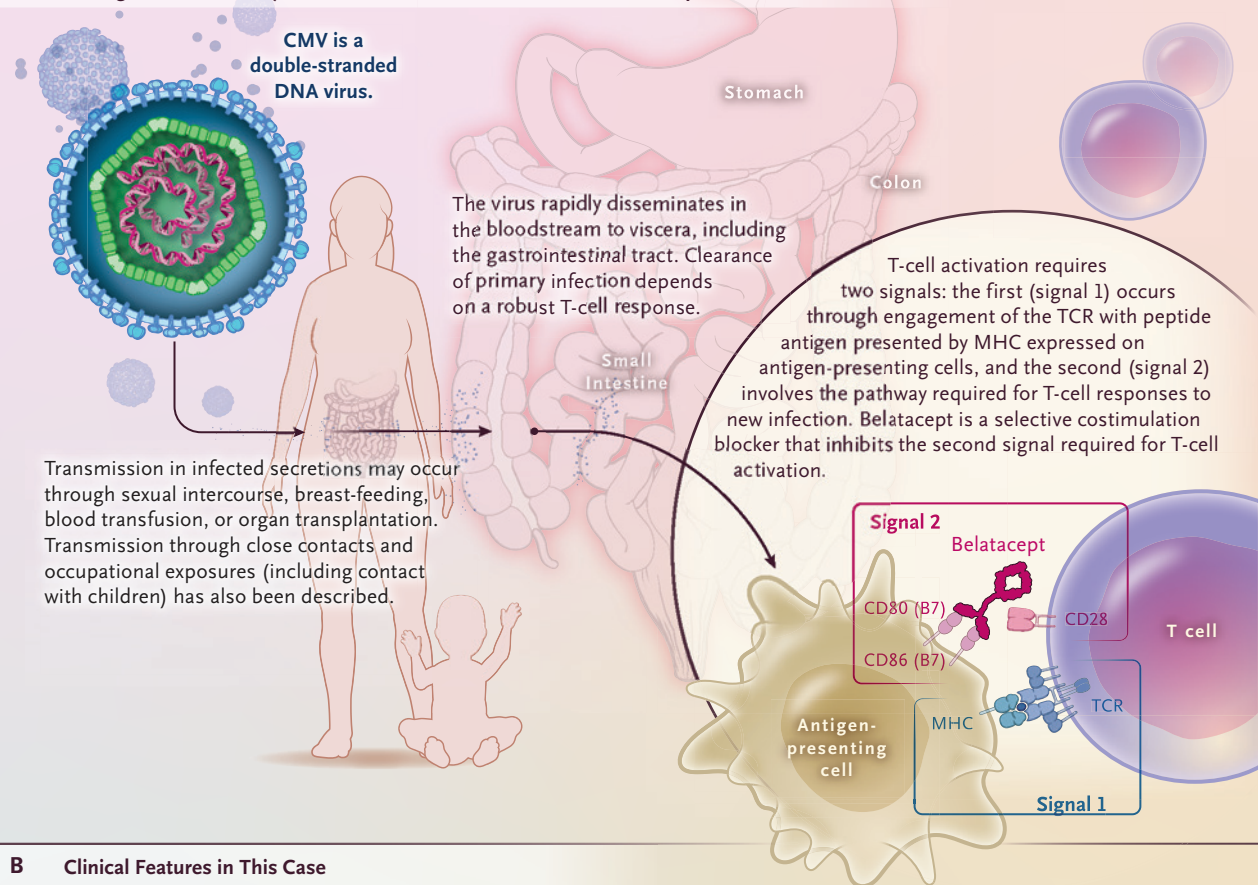
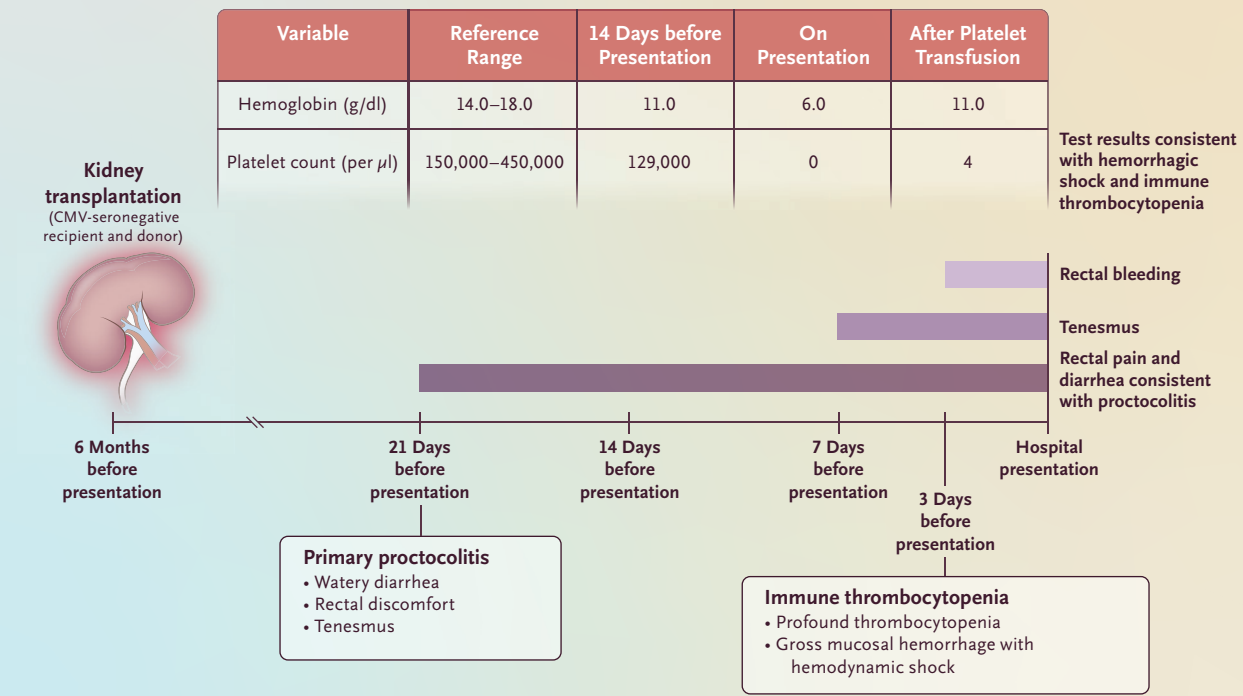
A Pathogenesis of Primary CMV Infection after Treatment with Belatacept**B Clinical Features in This Case**

Figure 2 (facing page). Primary Infection after Treatment with Belatacept.

Six months after kidney transplantation, this patient presented with a biphasic illness, characterized initially by proctocolitis and subsequently by immune thrombocytopenia. She had received belatacept, which is associated with impaired antiviral immunity, particularly in patients who have a primary viral infection and have not been previously infected with the virus. Panel A shows the pathogenesis of primary cytomegalovirus (CMV) infection after treatment with belatacept. MHC denotes major histocompatibility complex, and TCR T-cell receptor. Panel B reviews the clinical features that developed in this patient between the time of kidney transplantation and the current presentation, which are consistent with primary CMV infection.

readmitted to the hospital with a 1-day history of melena. Endoscopy revealed linear erosions; a biopsy specimen was not obtained. Treatment with high-dose glucocorticoids was considered to be a contributing factor, and a tapering course of prednisone was begun. Melena did not occur again during that admission. The CMV DNA viral load was 8000 IU per milliliter, and valganciclovir therapy was continued. Monitoring of the CMV DNA viral load was continued until suppression occurred, 4 weeks after the start of antiviral therapy. At multiple subsequent visits, the platelet count remained normal, at more than 150,000 per microliter, while the patient finished the prednisone course.

oral absorption.³⁶ Intravenous ganciclovir therapy was transitioned to oral valganciclovir therapy. The patient was discharged while receiving oral valganciclovir and high-dose prednisone, in addition to tacrolimus and mycophenolate. Outpatient endoscopy was planned.

Eight days after discharge, the patient was

FINAL DIAGNOSIS

Cytomegalovirus-induced immune thrombocytopenia.

This case was presented at the Medical Case Conference.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

REFERENCES

- Hamlyn E, Taylor C. Sexually transmitted proctitis. *Postgrad Med J* 2006;82:733-6.
- Siddiqui RS, Lakhdar S, Buttar C, Saliaj M. Thrombocytopenia secondary to herpes simplex virus-2 infection successfully treated by acyclovir. *Cureus* 2020;12(12):e12338.
- Kotton CN, Kumar D, Caliendo AM, et al. Updated international consensus guidelines on the management of cytomegalovirus in solid-organ transplantation. *Transplantation* 2013;96:333-60.
- Quinn TC, Goodell SE, Mkrtrichian E, et al. *Chlamydia trachomatis* proctitis. *N Engl J Med* 1981;305:195-200.
- Sigle GW, Kim R. Sexually transmitted proctitis. *Clin Colon Rectal Surg* 2015;28:70-8.
- Lavery EA, Coyle WJ. Herpes simplex virus and the alimentary tract. *Curr Gastroenterol Rep* 2008;10:417-23.
- Goodell SE, Quinn TC, Mkrtrichian E, Schuffler MD, Holmes KK, Corey L. Herpes simplex virus proctitis in homosexual men — clinical, sigmoidoscopic, and histopathological features. *N Engl J Med* 1983;308:868-71.
- Lee DH, Zuckerman RA. Herpes simplex virus infections in solid organ transplantation: guidelines from the American Society of Transplantation Infectious Diseases Community of Practice. *Clin Transplant* 2019;33(9):e13526.
- Whitaker JA III, Hardison JE. Severe thrombocytopenia after generalized herpes simplex virus-2 (HSV-2) infection. *South Med J* 1978;71:864-5.
- Kotton CN, Kumar D, Caliendo AM, et al. The third international consensus guidelines on the management of cytomegalovirus in solid-organ transplantation. *Transplantation* 2018;102:900-31.
- Jordan MC, Rousseau WE, Noble GR, Steward JA, Chin TD. Association of cervical cytomegaloviruses with venereal disease. *N Engl J Med* 1973;288:932-4.
- Pass RF, Little EA, Stagno S, Britt WJ, Alford CA. Young children as a probable source of maternal and congenital cytomegalovirus infection. *N Engl J Med* 1987;316:1366-70.
- Adler SP. Molecular epidemiology of cytomegalovirus: viral transmission among children attending a day care center, their parents, and caretakers. *J Pediatr* 1988;112:366-72.
- Angarone M, Snyderman DR. Diagnosis and management of diarrhea in solid-organ transplant recipients: guidelines from the American Society of Transplantation Infectious Diseases Community of Practice. *Clin Transplant* 2019;33(9):e13550.
- Lee C-Y, Chen Y-H, Lu P-L. Reactivated cytomegalovirus proctitis in an immunocompetent patient presenting as nosocomial diarrhea: a case report and literature review. *BMC Infect Dis* 2017;17:113.
- Studemeister A. Cytomegalovirus proctitis: a rare and disregarded sexually transmitted disease. *Sex Transm Dis* 2011;38:876-8.
- Shimanovsky A, Patel D, Wasser J. Refractory immune thrombocytopenic purpura and cytomegalovirus infection: a call for a change in the current guidelines. *Mediterr J Hematol Infect Dis* 2016;8(1):e2016010.
- Müller NF, Schampera M, Jahn G, Malek NP, Berg CP, Hamprecht K. Case report: severe cytomegalovirus primary infection in an immunocompetent adult with disseminated intravascular coagulation treated with valganciclovir. *BMC Infect Dis* 2016;16:19.
- Asranna A, Kumar A, Pranita, Goel A. Cold agglutinin mediated autoimmune hemolytic anemia due to acute cytomegalovirus infection in an immunocompetent adult. *Polish Annals of Medicine* 2016;23:43-5.
- Cines DB, Bussell JB, Liebman HA, Luning Prak ET. The TTP syndrome: pathogenic and clinical diversity. *Blood* 2009;113:6511-21.
- Yenicesu I, Yetgin S, Ozyürek E, Aslan D. Virus-associated immune thrombocytopenic purpura in childhood. *Pediatr Hematol Oncol* 2002;19:433-7.
- Schoettler M, Elisofon SA, Kim HB, et al. Treatment and outcomes of immune cytopenias following solid organ transplant in children. *Pediatr Blood Cancer* 2015;62:214-8.

23. Wei S-H, Ho M-C, Ni Y-H, Lin D-T, Lee P-H. Cytomegalovirus-associated immune thrombocytopenic purpura after liver transplantation. *J Formos Med Assoc* 2007;106:327-9.
24. Papagianni A, Economou M, Tsoutsou E, Athanassiou-Metaxa M. CMV-related immune thrombocytopenic purpura or CMV-induced thrombocytopenia? *Br J Haematol* 2010;149:454-5.
25. Sugioka T, Kubota Y, Wakayama K, Kimura S. Severe steroid-resistant thrombocytopenia secondary to cytomegalovirus infection in an immunocompetent adult. *Intern Med* 2012;51:1747-50.
26. Vincenti F, Charpentier B, Vanrenterghem Y, et al. A phase III study of belatacept-based immunosuppression regimens versus cyclosporine in renal transplant recipients (BENEFIT study). *Am J Transplant* 2010;10:535-46.
27. Xu H, Perez SD, Cheeseman J, Mehta AK, Kirk AD. The allo- and viral-specific immunosuppressive effect of belatacept, but not tacrolimus, attenuates with progressive T cell maturation. *Am J Transplant* 2014;14:319-32.
28. Chavarot N, Divard G, Scemla A, et al. Increased incidence and unusual presentations of CMV disease in kidney transplant recipients after conversion to belatacept. *Am J Transplant* 2020 December 6 (Epub ahead of print).
29. Karadkhele G, Hogan J, Magua W, et al. CMV high-risk status and posttransplant outcomes in kidney transplant recipients treated with belatacept. *Am J Transplant* 2021;21:208-21.
30. Bertrand D, Chavarot N, Gatault P, et al. Opportunistic infections after conversion to belatacept in kidney transplantation. *Nephrol Dial Transplant* 2020;35:336-45.
31. DiMaggio D, Anderson A, Bussel JB. Cytomegalovirus can make immune thrombocytopenic purpura refractory. *Br J Haematol* 2009;146:104-12.
32. Neunert C, Terrell DR, Arnold DM, et al. American Society of Hematology 2019 guidelines for immune thrombocytopenia. *Blood Adv* 2019;3:3829-66.
33. Emery VC, Griffiths PD. Prediction of cytomegalovirus load and resistance patterns after antiviral chemotherapy. *Proc Natl Acad Sci U S A* 2000;97:8039-44.
34. Roberts MB, Fishman JA. Immunosuppressive agents and infectious risk in transplantation: managing the “net state of immunosuppression.” *Clin Infect Dis* 2020 August 17 (Epub ahead of print).
35. Bate SL, Dollard SC, Cannon MJ. Cytomegalovirus seroprevalence in the United States: the national health and nutrition examination surveys, 1988-2004. *Clin Infect Dis* 2010;50:1439-47.
36. Asberg A, Humar A, Jardine AG, et al. Long-term outcomes of CMV disease treatment with valganciclovir versus IV ganciclovir in solid organ transplant recipients. *Am J Transplant* 2009;9:1205-13.

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